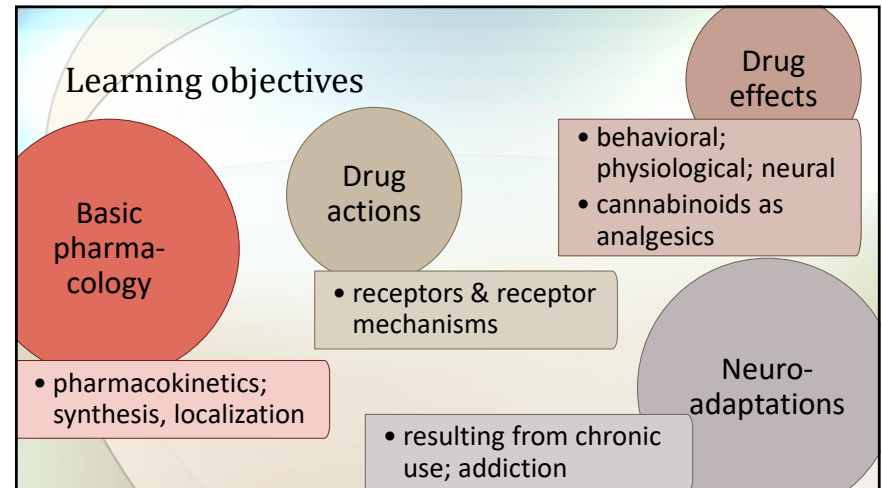
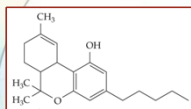


Cannabis and the Cannabinoids

Chapter 14

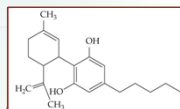


The psychoactive properties of cannabis result primarily from two *phyto*cannabinoids



Δ^9 -THC

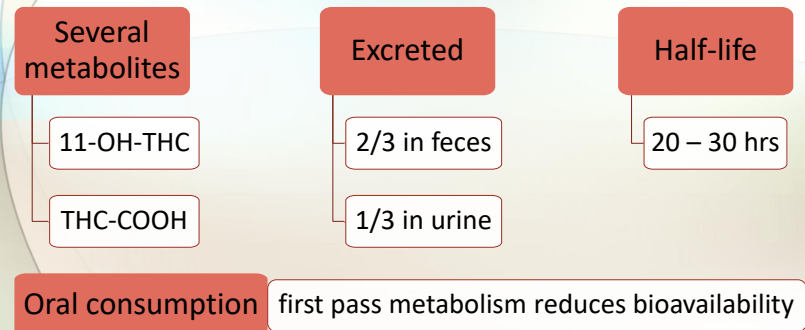
- inhaled or oral
- 20-30% absorbed into lungs
- lower bioavailability with oral consumption



Cannabidiol

- bioactive
- low affinity for dozens of targets
- therapeutic?

THC is metabolized by the liver, and distributed in adipose tissue



Chapter 14 – Cannabis & the Cannabinoids

There are 2 different cannabinoid receptors

CB₁

- primary CB receptor in the CNS
- wide distribution
 - cortex, basal ganglia, hippocampus, cerebellum
- presynaptic nerve terminal
 - $G_{i/o}$, $\uparrow g_K$, $\downarrow g_{Ca}$

- peripheral (bone, fat, GI)
- microglia
- low expression in neurons

CB₂

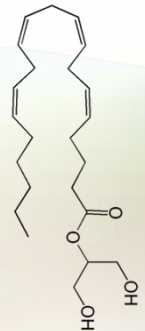
Endocannabinoids are fatty acid derivatives of membrane phospholipid.

Anandamide
arachidonoyl
ethanolamine (AEA)

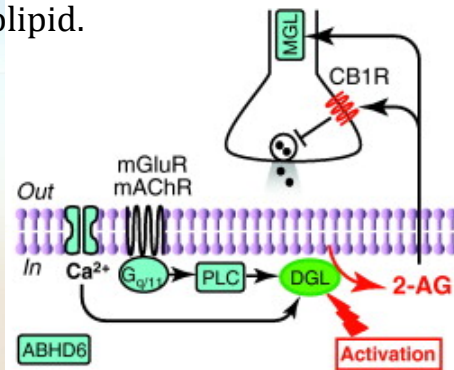
- CB₁ partial agonist
- multiple synthetic pathways
- degraded by FAAH
fatty acid amide hydrolase

2-AG
2-arachidonoylglycerol

- full agonist, nonselective
- CB₁ & CB₂
- degraded by MAGL
monoacyl-glycerol lipase



Endocannabinoids are fatty acid derivatives of membrane phospholipid.



Alger, B. E. & Kim, J. (2011). *Trends in Neurosciences*, 34, 304-315. DOI:10.1016/j.tins.2011.03.003

Endocannabinoids are fatty acid derivatives of membrane phospholipid.

DAG lipase (DGL) converts DAG → 2-AG

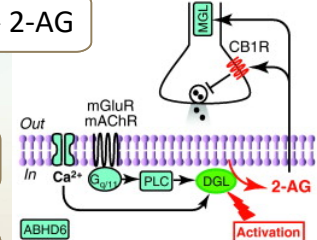
- $\uparrow [Ca^{2+}]$ by VGCC or NMDA-R
- GPCR / PLC (also PLA₂ & PLD)

monoacyl-glycerol lipase (MAGL)

- degrades 2-AG

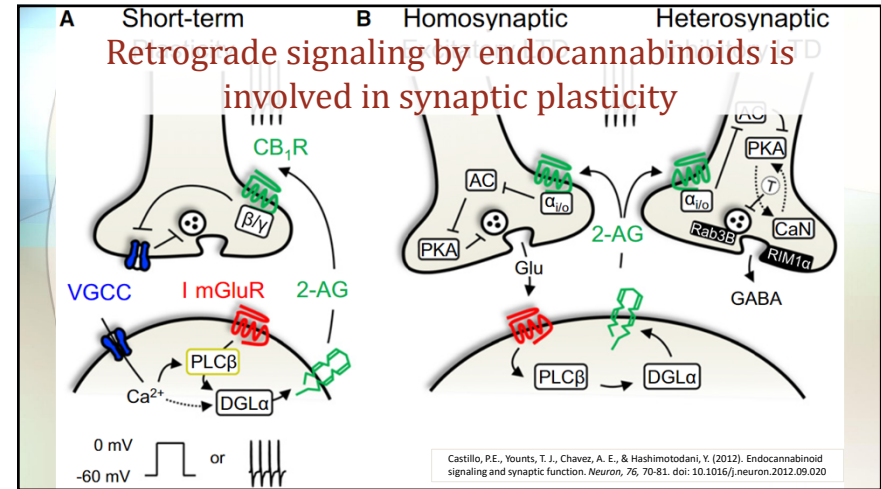
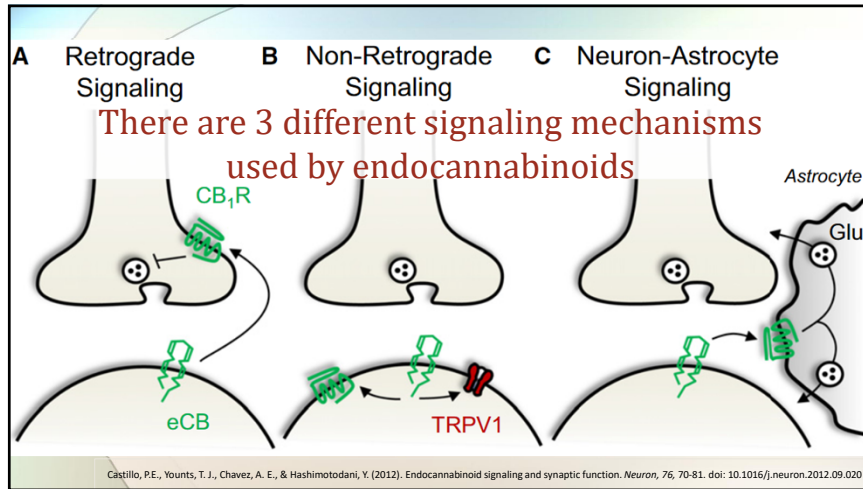
uptake transporter

- characterization is incomplete



Alger, B. E. & Kim, J. (2011). *Trends in Neurosciences*, 34, 304-315. DOI:10.1016/j.tins.2011.03.003

Chapter 14 – Cannabis & the Cannabinoids



Pharmacological and genetic techniques are used to determine the roles of CB receptors

Ligands

- *CP-55,940* & *WIN 55,212-2*
- full agonists CB₁ & CB₂
- *SR 141716A* (rimonabant)
- Selective CB₁ antagonist

Knock-out mice

- CB₁ KO
- CB₂ KO
- FAAH or MAGL KO

Learning objectives

Basic pharmacology

Drug actions

Drug effects

- behavioral; physiological; neural
- cannabinoids as analgesics

Neuro-adaptations

- resulting from chronic use; addiction

Chapter 14 – Cannabis & the Cannabinoids

CB₁ receptors mediate many of the effects of THC and endocannabinoids

Resultant effects of THC in mice

- decreased locomotion
- hypothermia
- catalepsy
- hypoalgesia

Functional effects of cannabinoids

- ↓ nausea/vomiting
- memory / extinction
- anxiolytic, antidepressant
- analgesic
- ↓ seizure

eCB signaling is anxiolytic and antidepressant

Humans report acute cannabis use produces feelings of relaxation, reduced anxiety and elevated mood.

Converging lines of evidence in rodents illustrate the anxiolytic and antidepressant effects of eCB signaling

Increase eCB signaling

Decrease eCB signaling

eCB signaling is anxiolytic and antidepressant

Increase eCB signaling

eCBs are anxiolytic in EPM, light-dark test

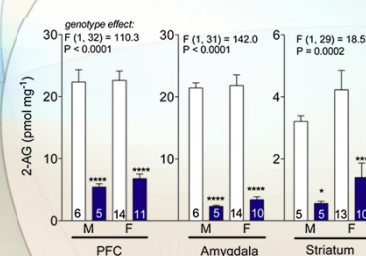
inhibition of FAAH prevents anxiety-like behaviors following social defeat stress

Decrease eCB signaling

CB1 KO show increased anxiety

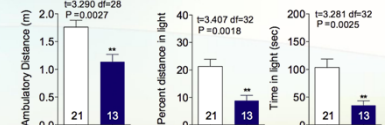
Mice with disrupted DGLα increases both anxiety & depression

DGLα deletion reduces 2-AG in PFC, amygdala and striatum

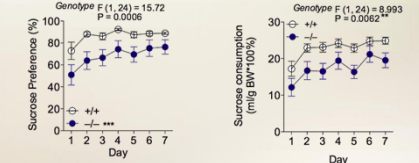


Shonesy, B.C., Bluett, R.J., Ramikie, T.S., Baldi, R., Hermanson, D.J., Kingsley, P.J., ... Patel, S. (2014). Genetic disruption of 2-arachidonoylglycerol synthesis reveals a key role for endocannabinoid signaling in anxiety modulation. *Cell Reports*, 9, 1644-1653. doi: 10.1016/j.celrep.2014.11.001

and increases anxiety



and increases anhedonia



Chapter 14 – Cannabis & the Cannabinoids

There is limited evidence that cannabinoids improve symptoms of anxiety or depression.

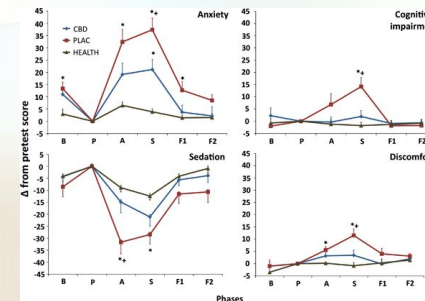
Depression

- No RCTs have specifically evaluated cannabis in patients with a depressive disorder.
- Evidence is inconclusive

Anxiety

- 4 RCTs examined anxiety symptoms in patients with chronic pain – 3 judged to have high risk of bias
- Only 1 RCT with high risk of bias (small N) found greater improvement with CBD on a simulated public speaking test

Anxiety improvement with CBD on a simulated public speaking test

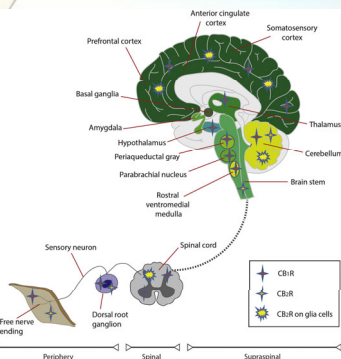


Bergamaschi, M. M., Queiroz, R. H., Chagas, M. H., Oliveira, D. C., Martinis, B. S., Kapczinski, F., ... Crippa, J. A. (2011). Cannabidiol reduces the anxiety induced by simulated public speaking in treatment-naïve social phobia patients. *Neuropsychopharmacology*, 36, 1219-1226. doi:10.1038/npp.2011.6

Hyperalgesia is induced by CB₁R or CB₂R KO, or chronic CB₁R antagonist SR141716A

Analgesic mechanisms

- inhibition of transmitter and peptide release
- modulation of postsynaptic neuron excitability
- activation of descending inhibitory pain pathway
- reduction of neural inflammation



Starowicz, K., & Finn, D. P. (2017). Cannabinoids and Pain: Sites and Mechanisms of Action. *Advances in Pharmacology*, 80, 437-475. doi:10.1016/j.pharm.2017.05.003

Vukobratovic, S., Srebro, D., Vukobratovic, K. S., Vukobratovic, C., & Prostran, M. (2018). Cannabinoids and pain: New insights from old molecules. *Frontiers in Pharmacology*, 9, 1253. doi:10.3389/fphar.2018.01253

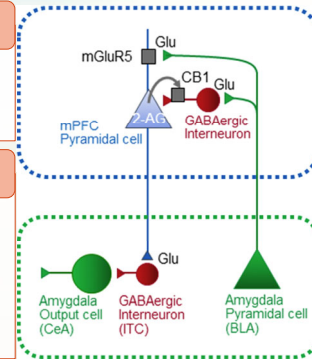
Cannabinoids affect the cognitive aspects of pain

mGluR5 in mPFC

- PLC → DAG → DAGLα → 2-AG synthesis

2-AG activates presynaptic CB₁

- ↓ GABA release, mPFC is disinhibited
- driving inhibitory neurons in the amygdala



Wozniak, S. G., Chappell, V., Finn, D. P., Chapman, A. G., & Neugebauer, V. (2017). The cannabinoid system and pain. *Neuropharmacology*, 124, 105-120. doi:10.1016/j.neuropharm.2017.06.015

Chapter 14 – Cannabis & the Cannabinoids

There is substantial evidence that cannabis is an effective treatment for chronic pain in adults

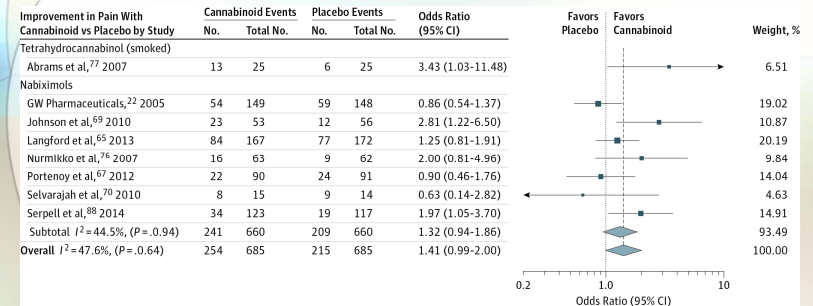
RCTs that compared cannabinoids to usual care, a placebo, or no treatment for 10 pain conditions

28 RCTs (2454 Ss): nabiximols (Sativex; 22),
smoke/vape (5), nabilone (5)

OR of reporting 30% or greater reduction of pain

Whiting, P. F., Wolff, R. F., Deshpande, S., Nisio, M. D., Duffy, S., Hernandez, A. V., ... Kleijnen, J. (2015). Cannabinoids for medical use. *Journal of the American Medical Association*, 313, 2456-2473. doi:10.1001/jama.2015.6358

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The first FDA-approved drug that contains a purified drug substance derived from cannabis



- FDA approved (2018)
- seizures associated with two rare and severe forms of epilepsy (age ≥ 2)
- Lennox-Gastaut syndrome
- Dravet syndrome

also the first FDA-approved drug for the treatment of Dravet syndrome.

Office of the Commissioner. (2018, June 25). Press Announcements - FDA approves first drug comprised of an active ingredient derived from marijuana to treat rare, severe forms of epilepsy. Retrieved March 28, 2019, from <https://www.fda.gov/NewsEvents/Newsroom/PressAnnouncements/ucm611046.htm>

The effect of THC

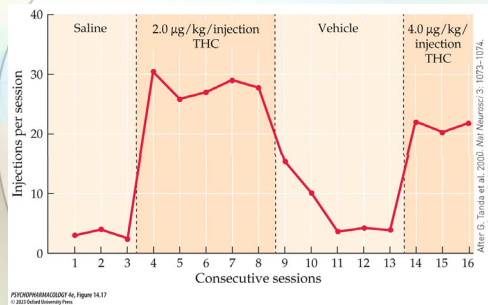


- Squirrel monkeys self-administer THC



Chapter 14 – Cannabis & the Cannabinoids

THC self-administration by squirrel monkeys



Role of CB₁

- blocked by SR141716A
- 2-AG is also self-administered

Cannabinoids are atypical drugs of abuse

self-administration

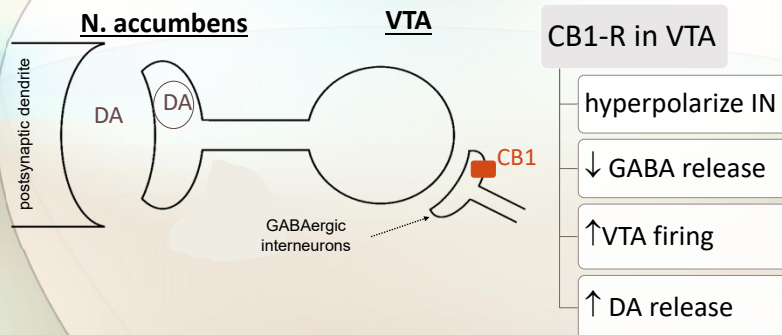
conditioned place preference

threshold of ICSS

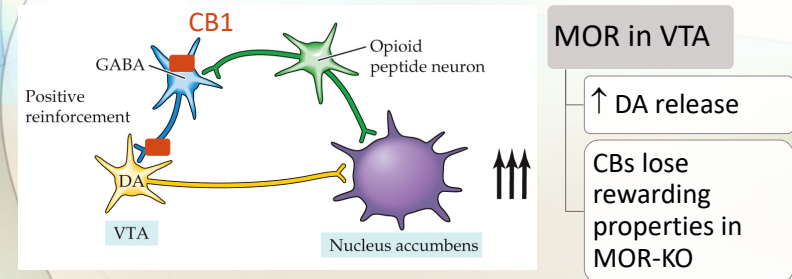
- only under certain conditions
- e.g. pre-exposed to drug, food or H₂O deprived, etc.

CBs are reinforcing under the appropriate experimental conditions & choice of species

Cannabinoids act on CB₁ receptors on GABAergic interneurons in the VTA to inhibit GABA release

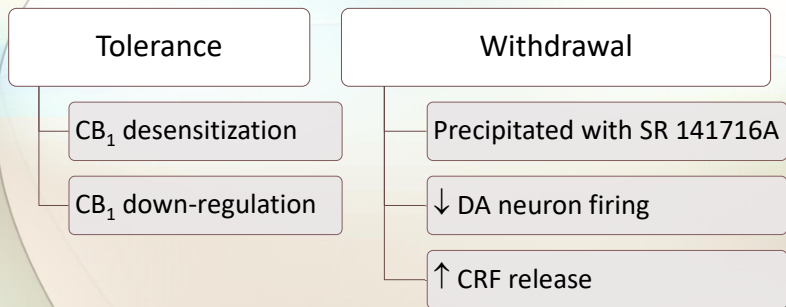


CB₁ and MOR receptors work together



Chapter 14 – Cannabis & the Cannabinoids

Chronic CB administration leads to tolerance and precedes precipitated withdrawal



Summary

- Behavioral effects of cannabinoids mediated largely through activation of CB₁ receptors
 - analgesic, anxiolytic, antidepressant & anticonvulsant effects
 - CB₁ receptors on GABAergic interneurons in VTA promote DA release
- Human clinical data generally supports therapeutic effects of cannabinoids only for pain & some seizure disorders
- Cannabinoids are atypical drugs of abuse,
 - reinforcing in animals only under certain conditions
 - but can show tolerance and withdrawal